

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1659

Cabinet Gorge 138 kV
August 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1659. The Interconnection Customer proposes a 61.3 MW solar photovoltaic facility interconnecting at the Cabinet Gorge 138 kV substation in the New Jersey area. The request was received on January 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 467 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,471,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Montclair Cogen Facility to Oleta 138.0 kV	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138.0 kV	163.4	186	8.4
Oleta Substation to Colorado 138.0 kV	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138.0 kV	161.1	216	8.4
Nietos to Colorado 138.0 kV	Colorado to Oleta 138.0 kV	130.0	172	8.8
Colorado Substation to Yahara River 138.0 kV	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138.0 kV	116.2	449	13.1
Spadra to Montclair Cogen Facility 138.0 kV	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138.0 kV	115.1	223	8.4
Lane to Passaic Valley Water Commission Substation 138.0 kV	Colorado to Yahara River 138.0 kV line	113.3	214	10.4
Elkhorn to Mckenna 138.0 kV	Colorado to Oleta 138.0 kV L	100.9	212	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oleta to Montclair Cogen Facility Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,342,000	12% (\$165,000)
Oleta to Colorado 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,160,000	12% (\$265,000)
Nietos to Colorado 138.0 kV	Rebuild line with higher-capacity conductor	\$824,000	18% (\$153,000)
Colorado to Yahara River 138.0 kV	Rebuild line with higher-capacity conductor	\$535,000	11% (\$58,000)
Montclair Cogen Facility to Spadra 138.0 kV	Rebuild line with higher-capacity conductor	\$1,604,000	12% (\$197,000)
Lane Substation to Passaic Vly Water Commission 138.0 kV	Rebuild line with higher-capacity conductor	\$3,040,000	20% (\$593,000)
Elkhorn to Mckenna 138.0 kV	Reconductor line	\$220,000	18% (\$40,000)
	Total allocated to project		\$1,471,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.